SmartAlarm — External Integration Standards and Deployment



Concept illustration — proposed integration, not an installed system.

Research/design baseline 3.0 · 24 October 2026 · English. Scope: physical fire protection integration, deployment engineering, and acceptance. This is a proposed integration design, not a certificate, installation approval, or substitute for the site fire-protection design. Manufacturer/model research is documented separately. The architecture below is a SmartAlarm proposal; cited standards summaries describe verified public scope rather than unverified clauses.

1. Decision and boundary

SmartAlarm shall support real physical fire-protection systems through approved interfaces while preserving autonomous local protection. It may ingest fire/supervisory/trouble states, coordinate responders, publish approved evacuation guidance, and request explicitly authorized actions. It must not require the cloud, Internet, a mobile app, or an LLM to complete the local automatic protective sequence.

Do not implement a universal “turn on sprinklers” API. Conventional heat-operated sprinkler heads act locally and individually; an interface to a fire panel does not imply that those heads are electrically addressable. Deluge, preaction, gas suppression, pumps, smoke control, and robotic monitors have different controllers, prerequisites, and consequences. Their permitted actions must be identified by system type and exact manufacturer configuration. The distinction between individual heat operation and software control is supported by the Pennsylvania Office of the State Fire Commissioner.

Three approvals remain separate: the applicable local code/edition and authority having jurisdiction (AHJ); the equipment listing/certification and manufacturer compatibility constraints; and acceptance of the complete installed integration. A protocol, a compliant component, or a cloud test does not establish all three. Record local adoption explicitly: an example official 2026 notice from Poudre Fire Authority cites earlier editions for specific inspection work, illustrating why the newest publication is not automatically the project edition.

2. Verified standards and interface matrix

Publication versions below were checked on 23 October 2026. The project must separately freeze adopted editions, amendments, equipment approvals, and contractual requirements. Exact paywalled clauses, installation dimensions, hydraulic calculations, and test intervals have not been inferred from public summaries.

Reference / verified version	Verified scope	Proposed SmartAlarm use and limit
NFPA 72, 2025	Fire alarm, signaling, emergency communication, and mass notification	Map alarm/supervisory/trouble and approved communication interfaces; no claim that SmartAlarm is a listed fire alarm panel. NFPA
NFPA 13, 2025	Design and installation of automatic sprinkler systems	Preserve the engineered sprinkler installation; collect waterflow/valve supervision through approved interfaces. Software does not replace sprinkler design. NFPA
NFPA 20, 2025	Selection and installation of stationary fire-protection pumps	Integrate approved controller status; any permitted remote request remains subordinate to the approved controller design. NFPA
NFPA 25, 2026	Inspection, testing, and maintenance of water-based fire protection	Track deficiencies, impairment, restoration, and evidence; schedules must be set from the adopted text and equipment instructions. Edition corroborated by NFPA's 2026 handbook and ANSI's NFPA 25 record.
EN 54-2 / -4 / -13	Control/indicating equipment; power supplies; component compatibility	Require exact certificates, configurations, and applicable national editions; no single "EN 54 certified system" shortcut. Public scope: BSI fire standards guide. Edition/amendment freeze remains a procurement task for these parts.
EN 54-16:2008 / EN 54-24:2008	Voice alarm control/indicating equipment; voice alarm loudspeakers	Use an approved voice alarm chain with zone mapping, priority, power/fault monitoring, and commissioned performance; an ordinary IP speaker is not automatically equivalent. BSI certification guide
ISO 7240-5:2018	Point-type heat-detector requirements and testing	Record detector class and approved application; not a universal environmental suitability claim. ISO
ISO 7240-16:2007 / -19:2007, both confirmed in 2025	Emergency sound control equipment; emergency sound-system design, installation, commissioning, and service	Distinguish component evidence from installed audibility/intelligibility and operating procedures. ISO part 16, ISO part 19
BACnet / ISO 16484-5:2026, edition 8, published August 2026	Building-system data communication and object-oriented representation	Require supported objects/services, revision, writable-property allowlist, priorities, and quality semantics. Interoperability does not authorize a life-safety action. ISO, BACnet committee
Modbus Application Protocol V1.1b3; Modbus Security	Transactions/function codes; TLS-based authentication and integrity for the security profile	Require exact register map, scaling, byte/word order, access mode, and supported security profile. Do not assume ordinary Modbus/TCP has Modbus Security. Modbus Organization
OPC UA / IEC 62541; verified part 2:2026	OPC UA security model	Require server namespace, node IDs, profiles, certificates, security policy, user rights, and status/timestamp semantics. Ordinary OPC UA use is not proof of a safety-certified control path. IEC

Reference / verified version	Verified scope	Proposed SmartAlarm use and limit
IEC 62443-3-2:2020; ISA/IEC 62443 series	IACS risk assessment, zones/conduits, target security levels; lifecycle security	Produce site risk assessment and zone/conduit controls. Do not assign a security level or claim certification merely because the gateway uses TLS. IEC, ISA
MQTT 5.0, OASIS Standard 7 March 2019	Publish/subscribe messaging	Edge telemetry transport with authentication, ACLs, expiry, deduplication, and bounded queues; QoS is not physical completion. OASIS
CloudEvents; deployed schema version must be pinned	Common event description/envelope	Carry source, type, event ID, time, and versioned domain payload; does not certify detection or actuation. CloudEvents
CAP 1.2, OASIS Standard 1 July 2010	Common alert exchange	Public advisories with source/update/cancel/area/expiry handling; never a direct suppression-release protocol. OASIS
MAVLink Command Protocol; vendor/flight-stack version pinned per aircraft	Command transport and command acknowledgments	Capability-specific gateway and pilot authority; ACK and physical outcome remain distinct. Protocol support grants no flight authorization. MAVLink
Aviation rules of the operating jurisdiction	Operator, aircraft, airspace, and mission authorization	Record jurisdiction, permission, pilot, flight constraints, and emergency procedure. The FAA public-safety guide is a United States example, not a worldwide permit.
WCAG 2.2	Web content accessibility	AA design target, with keyboard, focus, contrast, target, status, and text alternatives verified independently; does not establish acoustic coverage. W3C

Fire suppression media, special hazards, pressure equipment, electrical installation, smoke-control systems, and local fire/building law can add other standards. The site designer must identify these from the actual system; this matrix is not an exhaustive code list.

3. Proposed deployment architecture

```
flowchart TB
  CLOUD[SmartAlarm regional cloud: incident / policy / approval / audit]
  DMZ[OT DMZ: authenticated broker or reverse connection / firewall / monitoring]
  EDGE[Site edge: durable events / signed policy / command ledger / fencing]
  VG[Approved vendor gateway: supported API or supervised interface]
  LOCAL[Local approved fire / releasing / pump / voice controllers]
  DEV[Detectors / valves / pumps / suppression devices / speakers]
  MAN[Authorized local controls / manual stations / pilot]
  CLOUD <-->|events and bounded approved intents| DMZ
  DMZ <--> EDGE
  EDGE <--> VG
  VG <--> LOCAL
  LOCAL <--> DEV
  MAN --> LOCAL
  DEV -->|local initiating inputs| LOCAL
```

The local controller/device path runs independently of the cloud, DMZ, edge, identity service, and LLM. Existing approved hardwired/manual paths remain authoritative. SmartAlarm's edge is an integration component unless separately assessed and approved for another role; “edge” does not itself mean a safety controller.

Place cloud-facing services in an OT DMZ with explicit conduits to site systems. Prefer an authenticated site-initiated connection where supported; prohibit public Internet access to panel buses and general-purpose remote desktop into fire controllers. Separate telemetry ingestion from command authorization. Broker ACLs, firewalls, service identities, signed configurations, certificate rotation, and bounded maintenance access enforce that separation. These are proposed controls derived from the zone/conduit approach, not a claimed IEC 62443 certification.

Use a manufacturer-approved gateway or supervised I/O only where the exact design permits it. Specify input supervision, fault detection, isolation, wiring, electrical ratings, feedback, and commissioning evidence in the approved installation documents. No generic relay board bypasses a releasing panel, pump controller, interlock, or evacuation priority controller. Do not attach undocumented writes to a maintenance port. If only a common alarm contact exists, report common alarm—not a fabricated detector address or zone.

4. Integration permission tiers

Tier	Permitted behavior	Evidence required before enabling
T0 Inventory	Register assets/documents; no live connection	Site owner, exact models, location and ownership
T1 Observe	Read alarm, supervisory, trouble, flow, pressure, health, and status where available	Approved read path, point list, semantic mapping, FAT/SAT
T2 Operational coordination	Create incident/tasks; publish approved communications through the permitted voice interface	Approved message/zone/priority policy and verified delivery feedback
T3 Restricted control request	Only explicitly approved target/capability combinations	Manufacturer authorization, site engineering, AHJ/other acceptance as applicable, signed cause-and-effect revision, local interlocks, completion evidence
T4 Local automatic protection	Executes in the local approved protection system	Existing/local system design and acceptance; not granted to the cloud by a software entitlement

Default new connectors to T1 or lower. Alarm acknowledgment, silence, reset, disable/isolate, pump requests, and suppression release are distinct capabilities. Do not make a single generic “control” permission enable them all. Some remain local-only. An incident commander may coordinate the incident without possessing authority to reset or inhibit protection.

5. Asset, point, zone, and cause-and-effect contract

Each binding is versioned and reviewable before activation. Minimum contract fields:

Object	Required information
Asset	tenant_id, site_id, asset_id, controller_id, manufacturer, exact model, firmware, serial, controller/gateway relationship, protocol revision, certificate/listing reference and scope, maintenance owner, maintenance_state, impairment_ref
Point	point_id, controller_id, zone_id, raw address/object/node/register, type, read/write permissions, unit, scale, byte/word order if applicable, enumeration, normal/alarm/trouble/supervisory mapping, validity, observed_at, received_at, timestamp source, quality, stale_after
Zone	zone_id, physical boundary/floor, fire compartment reference, detection/suppression/voice zone types, point membership, exclusions, revision and approver

Object	Required information
Capability	capability_id, allowed target types, parameter schema/ranges, local preconditions/interlocks, manual/automatic authority, authorization_policy, command_ttl, expiry, idempotency_key, retry class, feedback_point_ids, completion criterion, unknown_outcome_policy
Cause and effect	revision, initiating condition, required local confirmations, effect, affected zone, priority, delay if approved, interlocks/inhibits, reset authority, failure response, test case and signoff
Evidence	source time, receive time, source quality, raw payload reference/hash, connector/firmware/config versions, operator and approval, command/result correlation

Never equate a location zone with a suppression zone automatically. A floor may have several control zones and a control zone may cross UI boundaries. Changes to device addressing or zoning invalidate the affected mapping and require review/test before write capabilities resume. Quality states are GOOD, BAD, STALE, UNKNOWN, and TEST; absence of data is not NORMAL.

A cause-and-effect record is the approved engineer/vendor/site document reflected in software, not a rule invented by an LLM. For example, a locally approved releasing sequence may report its initiation, pending state, interlock condition, and discharge feedback to SmartAlarm. SmartAlarm records and coordinates these states; it must not infer authorization to release from camera confidence or silence alarms as a side effect of closing an incident.

6. Command lifecycle and physical evidence

Proposed states: DRAFT → PENDING_APPROVAL → AUTHORIZED → SENT → ACCEPTED → EXECUTING → COMPLETED_VERIFIED; alternatives REJECTED, EXPIRED, FAILED, UNKNOWN, and RECONCILED. A controller may omit EXECUTING; the adapter must declare which states are observable. UNKNOWN is mandatory when the command may have arrived but the physical outcome is unverified.

Bind approval to command_id, target, capability, parameter digest, zone revision, cause-and-effect revision, requester, approver, writer_epoch, nonce, issued_at, and expires_at. Edge and vendor gateway enforce allowlists, freshness, authorization, and local controller acceptance. Local protection/interlocks take precedence. Mutating parameters or zone revisions invalidates approval. Failover fences the old writer; it never blindly repeats an uncertain physical action.

Record transport receipt, controller acceptance, and physical feedback separately. A pump-request ACK is not pressure/flow evidence; a voice-buffer ACK is not confirmed playback or intelligibility; a valve command is not confirmed position. Completion criteria are capability-specific and may require more than one instrument. Network timeout after transmission produces UNKNOWN, blocks automatic non-idempotent retry, and opens reconciliation with local staff/controller status. A software cancellation cannot promise to reverse released agent or a completed physical action. Reset and restoration follow a separate authorized process.

7. Failure-mode analysis and degraded operation

This is the initial qualitative FMEA; the site team assigns severity, likelihood, detectability, residual risk, and owners using site evidence. No generic numerical risk score is asserted.

Failure / consequence	Detection	Proposed containment and acceptance evidence
Cloud/Internet loss / remote visibility lost	Heartbeat age and connection state	Local automatic protection continues; edge buffers events; UI shows stale; prove with disconnection test
Edge loss / integration unavailable	Edge heartbeat and independent panel observation	Local sequence remains autonomous; cloud shows unavailable; restore from durable ledger without replaying uncertain writes; power-off edge test
Gateway restart or split brain / duplicate effect	Epoch/lease, durable command ledger	One fenced writer; uncertain writes reconciled, not replayed; reboot/failover tests
Wrong point or zone map / wrong action location	Independent point walkdown and revision digest	Disable affected write binding; verify every enabled target and negative cross-zone case
Stuck relay or failed feedback / false completion	Supervision and independent controller feedback where approved	No inferred success; local fault response and physical inspection under approved procedure
Lost power or depleted backup / incomplete coverage	Controller/power trouble and maintenance evidence	Preserve approved backup design; verify site-specific fail states and power restoration
Alarm storm / delayed critical event	Queue age, source rate, priority metrics	Reserve critical processing, bound queues, flag loss; local sequence never waits for broker
Compromised gateway / unauthorized request	Identity, allowlist, audit, anomaly monitoring	Revoke conduit/credential and isolate integration without disabling local protection
Maintenance mode confused with actual / unintended effect or ignored alarm	Explicit mode and local panel state	Separate test identities/routes, physical procedure, authorized test window, restoration checks
Clock skew or stale observation / unsafe premise	Source/receive times and quality flag	Block stale remote request; retain local protection; no timestamp rewrite hiding uncertainty
Controller ACCEPTED but no outcome / operator assumes success	Feedback deadline and missing independent evidence	UNKNOWN, local verification, alternative response; no blind retry
Voice zone unavailable / occupants not warned	Per-zone/controller fault and commissioned checks	Display coverage gap; initiate approved alternative communication procedure

8. Impairment, testing, and change governance

An impairment record contains affected assets/zones/functions, reason, start/end plan, responsible person, required notifications, compensatory measures determined by the site authority, access controls, and restoration proof. SmartAlarm does not decide that a degraded fire system is safe merely because alerts are acknowledged. Unplanned impairment escalates to the named site responsible person.

Test mode must have an explicit boundary. Software exercise labels alone cannot prevent a physical output from operating. Use only approved isolation/test arrangements, a responsible local technician, a reviewed test script, site notifications, and the required authorization. Simulator results do not authorize live agent discharge, pump operation, or emergency broadcasts. Live tests follow the adopted standard, vendor method, and local authority/site plan.

Configuration and firmware changes require impact assessment, signed version, rollback/recovery plan, and regression of affected cause-and-effect paths. No remote software rollback is assumed to restore a physical system state. Close impairment only after local restoration, normal supervisory state, target verification, outstanding-defect

review, and signed handback. Test frequency and compensatory measures remain site-specific; this document invents neither.

9. RFI, survey, FAT, SAT, and handover

Gate	Required deliverable	Exit evidence
RFI / procurement	Exact panel/controller/device models, firmware, regional variants, supported APIs/registers, read/write permissions, gateway licenses, listing/certificates, compatible accessories, lifecycle and maintenance support	Vendor written confirmation of each proposed capability; unresolved items stay disabled
Site survey	As-built drawings, fire strategy, hydraulic/system boundaries, cause-and-effect, zone/point schedule, communications/power, AHJ/adopted editions, operational roles and impairment process	Designer/site owner reviews mapping and identifies missing documentation
Detailed binding design	Network zones/conduits, point contracts, command state/feedback, local-only controls, approved interface/wiring, migration and rollback plans	Engineering/vendor/site approval and authority acceptance where applicable
FAT	Representative exact hardware/firmware or documented scope-limited simulator; positive/negative mappings, access control, bad quality, clock skew, duplicates, lost ACK, timeout, expiry, restart, failover, no-Internet operation	Signed results, raw evidence, defects resolved; simulator limitations explicit
SAT	Installed point-to-point verification, physical zone confirmation, approved cause-and-effect tests, local autonomy on loss of cloud/edge, supervised faults, voice coverage/performance per approved plan, restoration	Witnessed site evidence; every enabled command has verified result semantics; no unresolved critical defect
Handover / operation	As-built configuration, credential ownership, backup/restore, training, runbooks, maintenance schedule, impairment contacts, support boundaries	Site owner acceptance; inspection/authority records where required; ongoing change control

Acceptance means more than a green dashboard. For each enabled physical capability, demonstrate that a denied request cannot actuate, stale/expired requests cannot actuate, an uncertain outcome is shown as UNKNOWN, local controls retain priority, and cloud failure cannot stop the independent automatic sequence. Record response times measured in the approved test; do not claim a universal latency target without the site/controller acceptance criterion.

10. Evidence register and remaining engineering decisions

All links in section 2 and this register were consulted on **2026-10-23**. Evidence type is public official scope/catalog/technical guidance; none is a full installation conformity assessment.

Evidence group	Official sources	Confirmed / remaining limitation
E01 Fire alarm and suppression	NFPA 72, NFPA 13, NFPA 20, NFPA 25 handbook, ANSI NFPA 25	Titles/editions/scope verified; adopted clauses/amendments must be checked for the project. NFPA 13 scope was available in official indexed content; direct retrieval was intermittent.
E02 Detection/voice	BSI components, BSI voice, ISO 7240-5, ISO 7240-16, ISO 7240-19	Public part scopes; national EN editions/amendments and exact product certificates require procurement verification.
E03 Building/OT data	ISO BACnet, BACnet committee, Modbus, IEC OPC UA	Protocol scope/version; not vendor compatibility or safety approval.
E04 OT security	IEC 62443-3-2, ISA series	Risk/lifecycle framework; site target levels and evidence remain to be engineered.

Evidence group	Official sources	Confirmed / remaining limitation
E05 Events and warning	MQTT, CloudEvents, CAP	Messaging scope; no guarantee of physical completion.
E06 Drone/accessibility/local context	MAVLink, FAA, WCAG, Pennsylvania sprinkler explanation, Poudre adopted-edition example	Interface, jurisdiction example, accessibility target, and local physical context; not global compliance.

Open decisions before implementation: deployment countries and authorities; installed system types and exact variants; permission to read versus write each point; vendor gateway and certificate scope; approved cause-and-effect; local-only actions; physical completion feedback; cybersecurity risk assessment; site-specific acceptance criteria and responsible signatories. These are required engineering inputs, not reasons to replace the full physical integration scope with a demo.

UX2-PHYSICAL · Visual supervision and physical feedback

The v2 console uses a pipe/valve/pump mimic, zone coverage and explicit device state. A requested command, gateway acknowledgement and observed physical effect are separate states. Unknown or stale feedback must use a labeled pattern and timestamp, never a green success state. Review-only actions in the prototype do not operate equipment. Local fire protection authority, interlocks and commissioning requirements in this document remain binding.

Verify F08–F20 with communication loss, conflicting local state and delayed feedback. Operator access to the schematic must not bypass command authorization. PA playback feedback is not proof of audibility to every occupant. Drone/robot task acceptance is not proof of arrival or rescue.

See UI/UX design, screen specifications and redesign audit.

11. BMS integration deployment contract — site implementation

Engineering bundle and commissioning sequence

1. **Survey:** inventory supervisors, servers/engines/controllers, product/build/firmware, API/module/provider/license, protocol/PICS, point/alarm dictionaries, network routes/ports, principal scopes, local emergency authority and maintenance windows. Record unknowns explicitly; unknown capabilities remain disabled.
2. **Read-only qualification:** configure endpoint allowlists, secret references/certificates, clocks and measured budgets. Discovery is bounded to approved roots/scope, cancellable and logged. Verify authentication, pagination, subscriptions/polling, native timestamps, quality, reconnect and active-alarm reconciliation before production admission.
3. **Mapping qualification:** stage and run golden fixtures for raw addresses, types, endianness, scale/offset, units, enums, severity ordering, native occurrences and ACK tokens. Map equipment/room/HVAC/fire-compartment/route/PA groups with typed relationships; verify direct/mirror lineage. Publish revisions with impact graphs and independent review.
4. **Runtime admission:** registry records adapter binary/hash, profile/mapping versions, qualified build/license and site acceptance. Servers calculate effective capabilities from current evidence. Exercises use separate namespaces and connector sessions; UI switches cannot enable writes.

5. **Site SAT:** measure actual load/bursts and clock quality; execute BMS-T01–20/BMS-P-01–28 per capability. Local protection/override remains cloud-independent. Record deviations, rollback/handover, owners and signed acceptance; reads/history/ACK/commands/release receive separate sign-off.
6. **Handover:** deliver data-freshness dashboards, alarm-gap runbooks, certificate/license expiry, support scopes, backup/restore, upgrade/cutover, priority-ownership cleanup and retirement. Update inventory and evidence when builds or equipment change.

Network and security

Site edges terminate BMS/OT communication in approved segments; northbound communication crosses authenticated OT-DMZ conduits with tenant/site binding and restricted egress. Do not expose BACnet/Modbus/OPC DA directly to the Internet. BACnet/SC, OPC UA security or Modbus Security requires correctly supported endpoint profiles; legacy TCP is not implicitly encrypted. OPC DA adapters require a site-approved compatible Windows/DCOM bridge/runtime and cannot be relabeled OPC UA. Secrets reference a vault; audits/exports remove tokens and clinical data. Apply NIST OT segmentation and site risk assessment without treating a standard as whole-system certification. NIST SP800-82r3.

Protocol-specific contract

Interface	Semantics to verify	Prohibited inference
BACnet IP/MS-TP/SC	Objects/properties, PICS, COV lifetime, notification/ACK, status flags, priority arrays and relinquish default	WriteProperty does not grant life-safety authority; application TTL cannot clear retained slots automatically; cloud leases do not prove slot ownership
OPC UA	Namespace URI, stable NodeIds, status/timestamps, security policy, A&C occurrence/branch/EventId, ConditionRefresh and history capability	Subscription alone does not load all active alarms; refresh is not replay; DA/A&E/HDA are not UA
Modbus	Register maps, function codes, zero/one-based addresses, byte/word order, scale, quality provenance and polling budgets	Register changes do not create native alarm lifecycle/ACK; plain TCP is not Modbus Security
Native REST/SOAP/SSE	Authentication/session expiry, pagination/tokens, object IDs, event semantics, per-method scopes/licenses, partial batch failures and restart cursors	HTTP200 cannot confirm every batch item; event ACK need not change whole-alarm state; outbound webhooks do not establish inbound command APIs

Recovery, failover and prohibited shortcuts

Lost source cursors or restarts move connections to reconciling. Commit active-snapshot completion markers; partial snapshots cannot clear missing alarms. Replay/history carries mode and cannot create broadcasts/commands. Duplicates retain receipts/canonical lineage; incident-writeback echoes are isolated before intake rules. Edge journals are durable before acceptance; disk pressure generates backpressure/gap alarms, not silent discard. Write fencing must operate at real egress; if unproven, standby remains observe-only with operator recovery. Local override revokes remote eligibility; reconnect never restores old setpoints automatically. BACnet slots retained after power loss need site-approved local watchdogs or cleanup; cloud TTL alone cannot guarantee release.

Acceptance evidence and operating contract

Each case records test revision, deployment fingerprint, input fixtures/expected observations, raw native statuses/timestamps, policy/approval results, redacted transport evidence, any physical witnesses, result and exceptions. Measure source→edge, edge commit→cloud, cloud→operator, request→native receipt and receipt→independent feedback separately. Freshness thresholds differ from latency SLAs and depend on hazard/site;

historian backfills cannot starve alarm queues. Certificate/license rotation measures interruptions and reconciles data rather than suppressing errors to report healthy. Invalidate admission evidence for the affected scope while retaining unrelated capabilities only where evidence still holds.

Sources and implementation artifacts

The existing Physical Device Catalog incorporates 11 source-backed vendor profiles. FRD defines FR-BMS-01–40/WF-34–37. Basic Design and Detail Design are the primary implementation specifications; the BMS source ledger retains 45 sources and research limitations. Engineering JSON profiles are not runtime authorization; activation requires site-specific evidence and the corresponding runtime policy/schema.